

# Sajeevan Veeriah

## Robotics, Automation, Control Systems and Embedded Systems Engineer

+61 498 586 654 | sajeevanveeriah@gmail.com | linkedin.com/in/sajeevan-veeriah | sajeevanveeriah.github.io  
Graduate Member - Engineers Australia

### PROFESSIONAL SUMMARY

Mechatronics engineer focused on robotics, industrial automation, control systems, embedded systems, IoT, validation and applied AI/ML. Practical background across smart factory automation, HMI/SCADA/MES, CAN-based vehicle software validation, instrumentation, data acquisition, manufacturing operations, CAD, QA records and technical documentation. Strong at connecting sensors, actuators, embedded devices, control logic, data pipelines and operator workflows into testable, reliable systems.

### CORE ENGINEERING POSITIONING

- Systems integration across electrical, mechanical, embedded, software, data and controls layers, with emphasis on fault isolation, commissioning support, traceability and handover readiness.
- Robotics and autonomy foundation across ROS 2, Gazebo, RViz, SLAM, A\* path planning, Kalman/EKF state estimation, sensor fusion, LiDAR/IMU concepts, OpenCV and PID control.
- Automation and regulated manufacturing exposure across functional design, FAT, SAT, qualification, process visualisation, MES, batch execution, diagnostics and maintainability.

### TECHNICAL SKILLS

**Controls and Automation:** PLC fundamentals, Siemens TIA Portal, WinCC, iFIX, PVI+, HMI, SCADA, MES, batch execution, Modbus TCP/RTU, Profinet, industrial Ethernet, FAT, SAT, commissioning support and troubleshooting.

**Robotics and Autonomy:** ROS 2, Gazebo, RViz, SLAM, A\* planning, Kalman/EKF concepts, state estimation, sensor fusion, LiDAR, IMU, depth camera concepts, OpenCV, motion control and PID tuning.

**Embedded and Electronics:** ESP32, STM32 concepts, embedded C/C++, Python, FreeRTOS concepts, UART, I2C, SPI, CAN, ADC, PWM, sensors, actuators, Altium, KiCad and signal conditioning.

**Software, Data and AI/ML:** Python, C, C++, MATLAB, Simulink, JavaScript, Node.js, REST APIs, Git, Linux, JIRA, test automation, InfluxDB, Grafana, data analysis, ML fundamentals and anomaly detection concepts.

**CAD, Manufacturing and Quality:** SolidWorks, Fusion 360, mechanical integration, drawings, 3D printing, FEA exposure, VFD exposure, pneumatics, hydraulics, ADR, EURO, ISO/IEC/AS/NZS, ITPs, MDRs, FMEA and QA records.

### PROFESSIONAL EXPERIENCE

#### Automation Engineer | JAG Process Solutions Pty Ltd

Brunswick, VIC | Jan 2026 - Jun 2026

- Worked on automation and smart factory systems for pharmaceutical, biotech, food and process clients where reliability, traceability, validation discipline and maintainability were central requirements.
- Supported control logic, HMIs, process visualisation, MES, batch execution and production reporting data across automated plants, skids and package units.
- Contributed to requirements review, functional design, FAT, SAT, commissioning, qualification, troubleshooting and handover documentation activities.
- Integrated field devices, sensors, drives and process equipment with HMI, control logic, production data and workflows while considering diagnostics, safety, usability and data integrity.

#### Product Development Test Engineer | Ford Motor Company via Invenio

Little River, VIC | Oct 2025 - Jan 2026

- Worked in the Feature Software Integration and Validation team on vehicle software integration and ADAS validation for T6 Ranger and Everest programmes.
- Executed feature vehicle, breadboard and regression testing for readiness, running changes, OTA updates and sign-off evidence across vehicle variants.
- Handled vehicle instrumentation and structured test drives using corporate test procedures in controlled and real-world conditions.
- Used Vector CANoe, CANalyzer and diagnostic tools to capture and analyse CAN bus data for fault isolation, verification and evidence-based defect reporting.

#### Technical Assistant | ABMARC

South Geelong, VIC | Jul 2024 - Aug 2025

- Supported vehicle emissions and compliance testing against Australian Design Rules and EURO standards across multiple vehicle platforms.
- Operated emissions test equipment, instrumentation and data acquisition systems with focus on repeatable, auditable and defensible results.
- Prepared compliance, QA and regulatory documentation for certification, audits and reporting while analysing test data to identify deviations and trends.

## PROFESSIONAL EXPERIENCE CONTINUED

### Consultant Engineer - Internet of Things | DuxTel Pty Ltd

Greater Geelong, VIC | May 2024 - Aug 2024

- Designed and deployed IoT solutions linking sensors, embedded devices, gateways, cloud platforms, data pipelines and user-facing dashboards.
- Built real-time data ingestion, monitoring and visualisation using InfluxDB and Grafana, with documentation and post-deployment support for handover.

### IoT Projects Administrator | DuxTel Pty Ltd

Greater Geelong, VIC | Feb 2024 - May 2024

- Configured IoT devices into ChirpStack-based LoRaWAN networks and developed an ESP32-based GPS tracking system with environmental sensing.
- Supported firmware configuration, device provisioning, testing, deployment workflows and demonstration of smart city, asset tracking and safety concepts.

### Production Worker and Cellar Hand | IDL

Moorabool, VIC | Aug 2022 - Feb 2024

- Ran and monitored food and beverage production equipment, maintained throughput and quality, completed checks, reported faults and supported traceability.

### Production Line Operator | Carbon Revolution

Waurm Ponds, VIC | Sep 2021 - Dec 2021

- Set up and operated manufacturing equipment for high-performance carbon fibre wheels while maintaining quality, safety, routine maintenance and first-level troubleshooting requirements.

### Undergraduate Engineer | Thornton Engineering Australia Pty Ltd

Corio, VIC | Mar 2018 - Mar 2020

- Supported engineering and QA for structural steel fabrication delivered to Australian Standards and client specifications.
- Prepared Inspection Test Plans, Manufacturer Data Reports and quality records, and reviewed drawings for inspection planning, material traceability and compliance.

## SELECTED ENGINEERING PROJECTS

- Autonomous Navigation and ROS 2 Robotics: ROS 2, Gazebo, RViz, A\* path planning, SLAM, Kalman/EKF-based state estimation, odometry and IMU fusion.
- ESP32 Clinical Ataxia Assessment Concept: ESP32 embedded measurement concept using IMU, ToF, Hall-effect and magnetometer sensing with firmware logic and data acquisition.
- IoT GPS and Environmental Monitoring Platform: ESP32 GPS and environmental sensing with LoRaWAN/MQTT telemetry, device configuration, InfluxDB/Grafana dashboards and deployment documentation.
- Smart Factory and Process Visualisation: HMI/process visualisation, MES and batch-system concepts where usability, traceability, diagnostics and maintainability are engineered together.

## PROJECT DELIVERY AND OPERATING STRENGTHS

- Requirements interpretation, scope breakdown, issue escalation, task tracking, test evidence, documentation control and handover readiness across engineering and production settings.
- Cross-functional collaboration with automation, mechanical, electrical, firmware, data, QA, production and customer-facing stakeholders.
- Technical documentation across test plans, defect records, QA records, Inspection Test Plans, Manufacturer Data Reports, commissioning records, validation evidence and operating handover notes.
- Practical delivery mindset across office, workshop, lab, production floor, proving ground and field environments with strong safety, quality and traceability discipline.

## EDUCATION, MEMBERSHIPS AND CERTIFICATIONS

### Bachelor of Mechatronics Engineering (Honours)

Deakin University | VIC | Mar 2018 - Jul 2025

### Higher National Diploma, Mechatronics, Robotics and Automation Engineering

Cardiff Metropolitan University | 2014 - 2016

Member, Engineers Australia | Current Victorian Driver Licence | Lean Six Sigma Foundation | JIRA and Agile | KAIZEN | LinkedIn Learning: Industrial Automation and IoT Foundations

Languages: English, Tamil and Sinhala

## MEMBERSHIPS AND CERTIFICATIONS

Member, Engineers Australia

Current Victorian Driver Licence

Lean Six Sigma Foundation

JIRA and Agile

KAIZEN

**Languages:** English, Tamil and Sinhala